Molecular dynamics of TEM-1 β-lactamase with six ligands

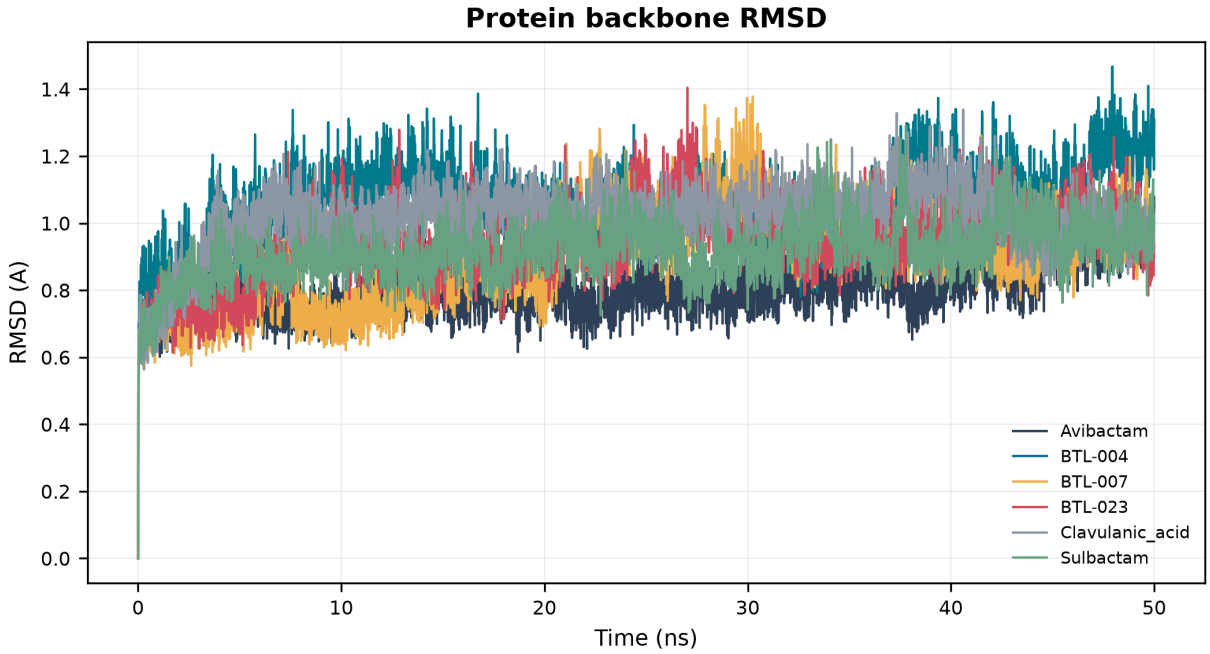
OpenMM, Amber ff14SB + OpenFF Sage 2.1.0, AM1-BCC charges, TIP3P water, 0.15 M NaCl, 300 K, 1 bar, 4 fs timestep with hydrogen mass repartitioning. Ligands protonated for pH 7.4. Starting poses are the top-scoring AutoDock Vina poses, unchanged.

Run summary

Compound	Protonation	Charge	ns	BB RMSD	Lig RMSD	COM dist	H-bonds >30%
Avibactam	sulfate/sulfonate	-1	50.0	0.84	2.2	4.36	3
BTL-004	tetrazolate	-1	50.0	1.08	5.14	7.3	3
BTL-007	neutral	0	50.0	0.99	28.27	27.11	0
BTL-023	neutral	0	50.0	1.0	33.57	28.25	0
Clavulanic_acid	carboxylate	-1	50.0	1.04	1.27	4.42	4
Sulbactam	carboxylate	-1	50.0	0.95	6.17	9.6	1

Mean values are taken over the second half of each run, after equilibration. 'COM dist' is the ligand centre of mass to the catalytic centroid.

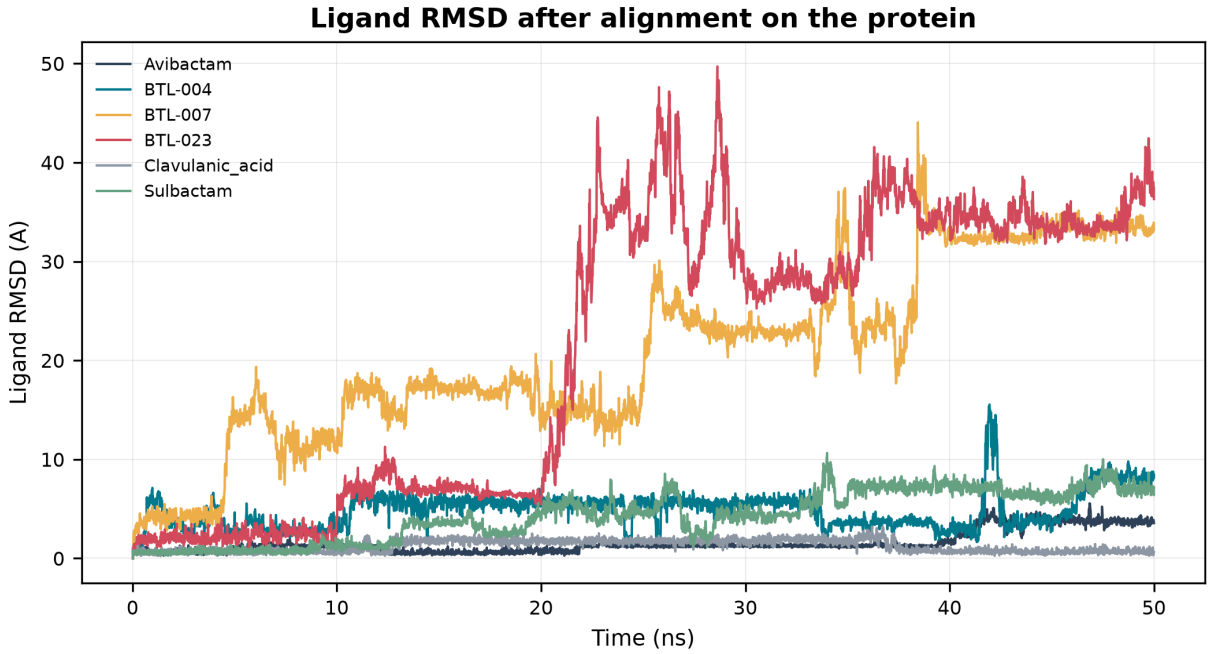
Figure 1. Protein backbone RMSD



File: figures/backbone_rmsd.png

How far the protein backbone has moved from its starting structure at each point in the run. It should rise over the first few nanoseconds and then flatten. A plateau around 1-3 A means the enzyme is stable and everything downstream can be trusted. A line still climbing at the end means the system never equilibrated.

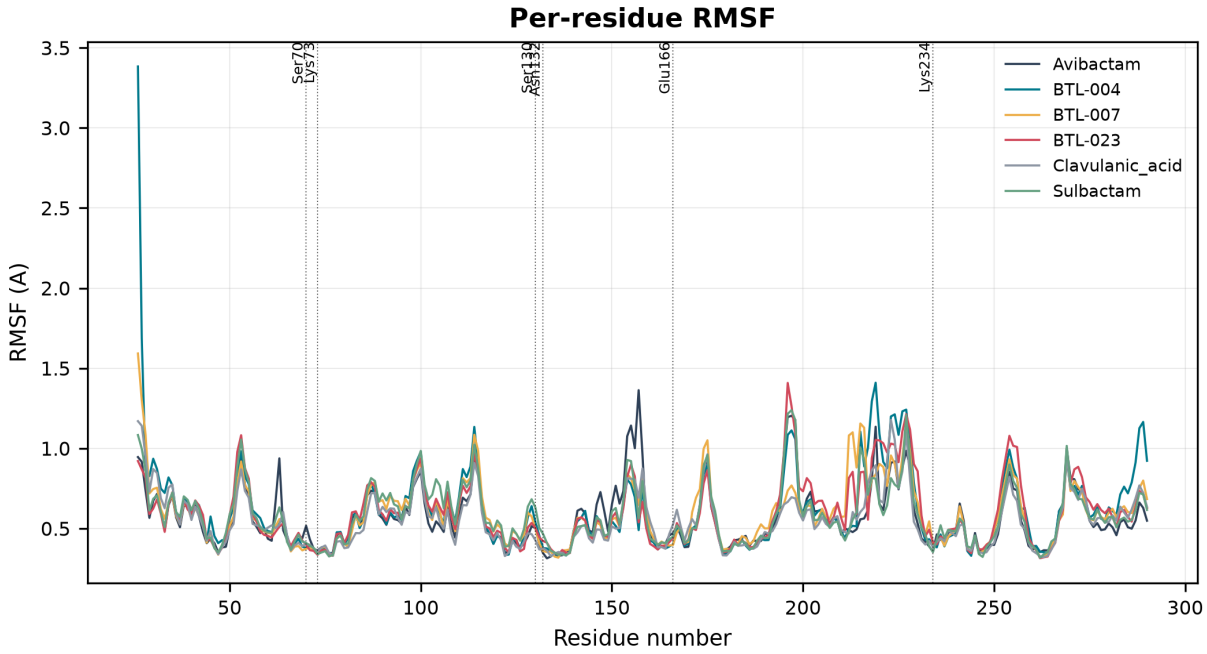
Figure 2. Ligand RMSD after alignment on the protein



File: figures/ligand_rmsd.png

The most important plot here. The protein is superposed frame by frame, so this measures how far the compound moved RELATIVE TO THE POCKET. Low and flat means it stayed where it was docked. A step change means it slid or flipped into a different pose. A steady climb means it left the site. Backbone RMSD alone cannot tell you any of this.

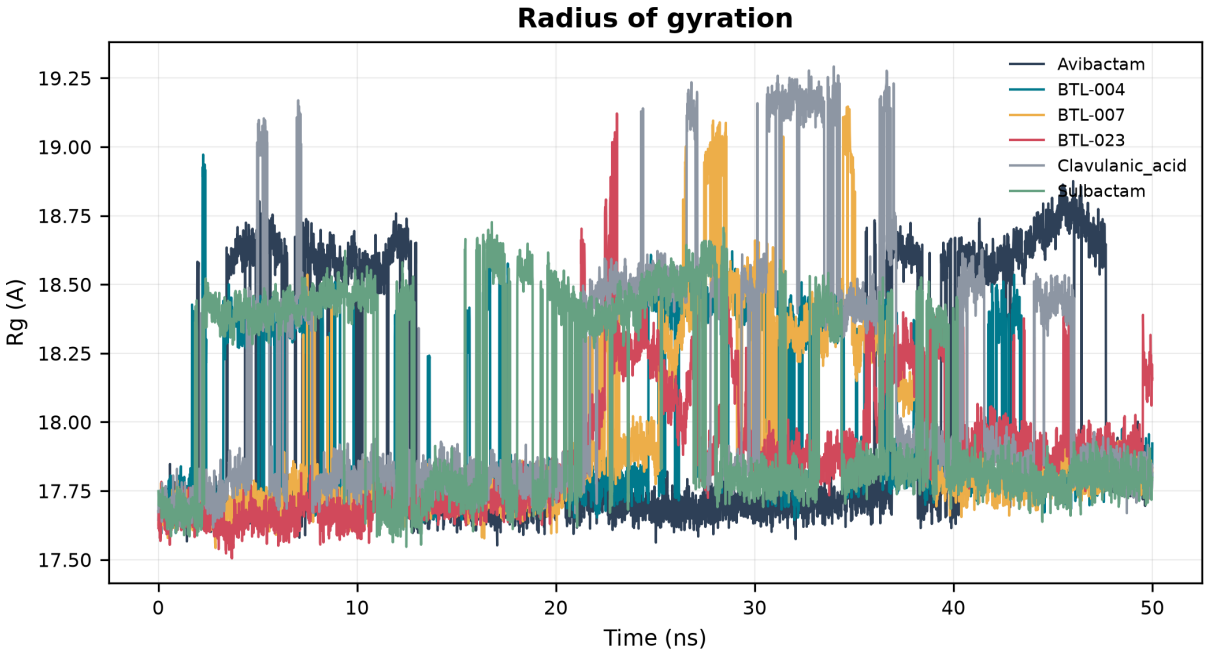
Figure 3. Per-residue RMSF



File: figures/rmsf.png

Average fluctuation of each residue over the run. Loops are flexible and spike; the core is rigid and low. The six catalytic residues are marked. If a catalytic residue is less mobile with one compound bound than with another, that compound is holding the site more rigidly.

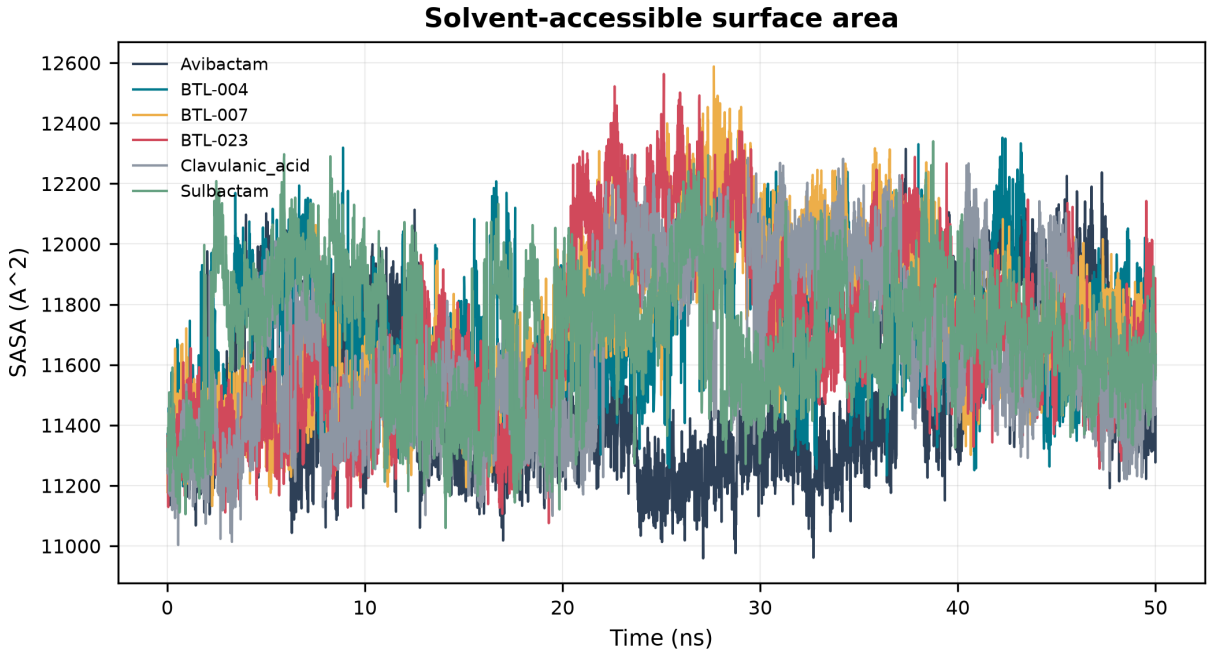
Figure 4. Radius of gyration



File: figures/rgyr.png

How compact the protein is. It should be an essentially flat line. A rising trend would indicate unfolding, which for a 50 ns run at 300 K would point to a setup problem rather than real biology.

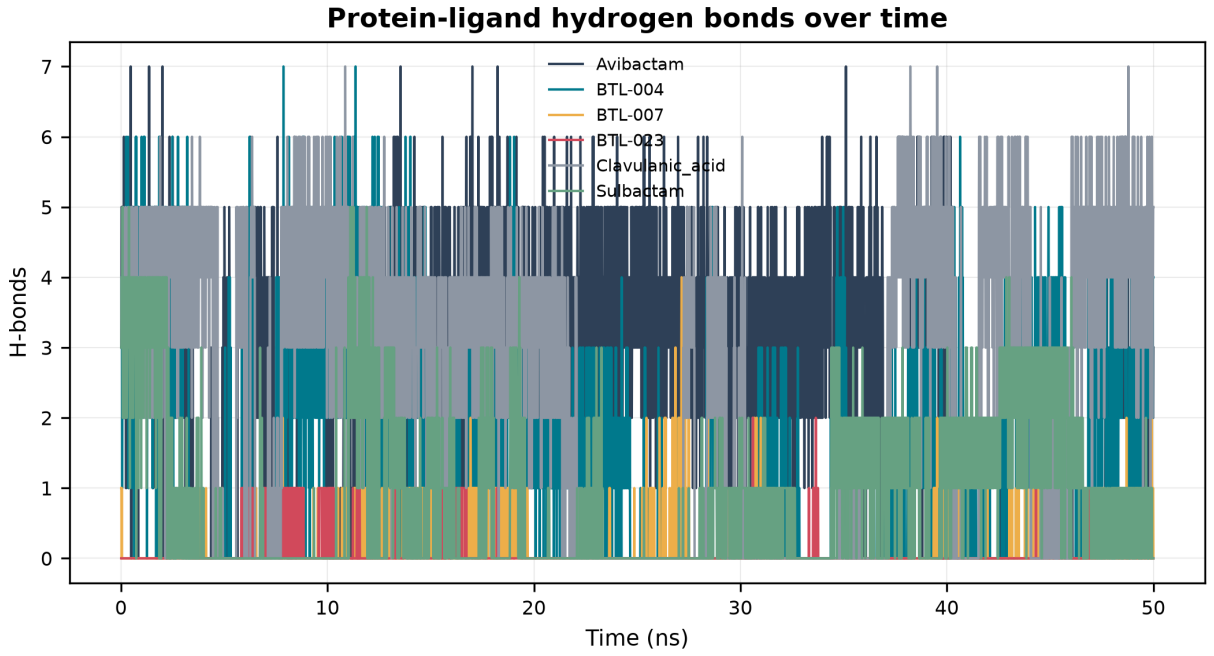
Figure 5. Solvent-accessible surface area



File: figures/sasa.png

How much of the protein surface is exposed to water. Also expected to be flat. A modest drop can reflect the pocket closing around the ligand.

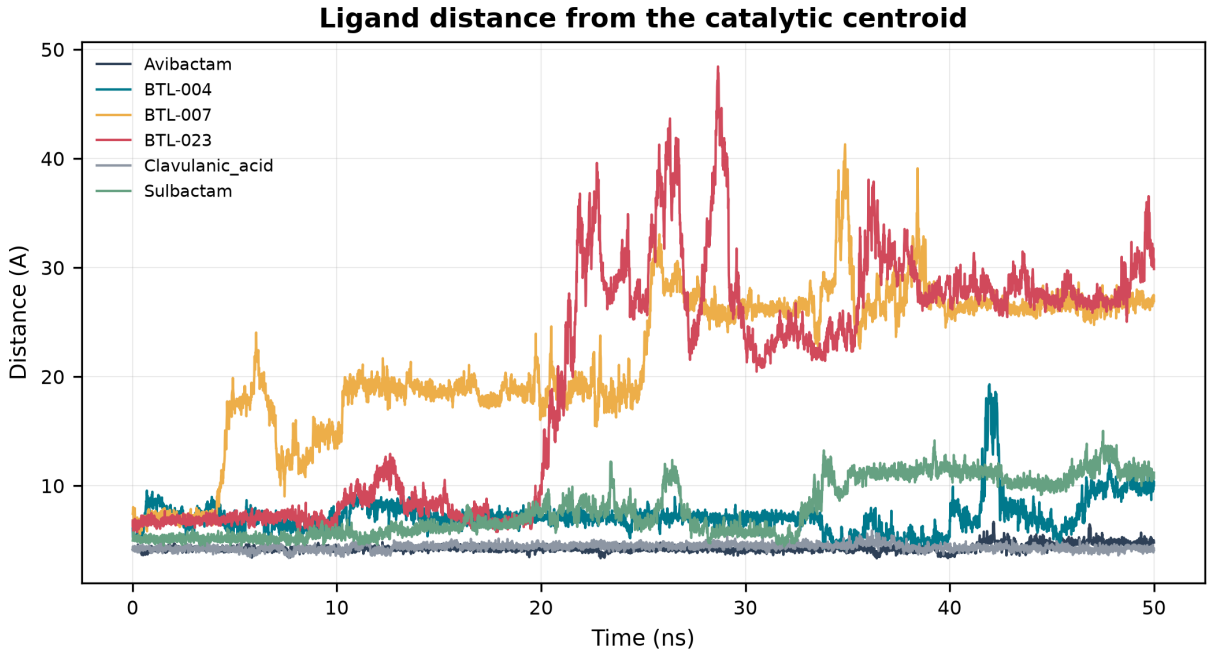
Figure 6. Protein-ligand hydrogen bonds over time



File: figures/hbonds.png

The number of hydrogen bonds between the compound and the enzyme at each frame. The running count is noisy by nature; the occupancy table that follows is the quantitative version and is what belongs in a manuscript.

Figure 7. Ligand distance from the catalytic centroid



File: figures/com_dist.png

Straight-line distance from the compound's centre of mass to the centroid of the six catalytic atoms (Ser70 OG, Lys73 NZ, Ser130 OG, Asn132 ND2, Glu166 OE1, Lys234 NZ), recomputed every frame from the protein itself and measured with the minimum-image convention so a ligand crossing the periodic boundary is not counted as having left. Flat and small means bound; a rise means the compound is leaving. This is the bluntest and hardest-to-misread measure of whether a pose survived.

Hydrogen-bond occupancy

Percentage of production frames in which each protein residue formed a hydrogen bond with the compound (donor-acceptor ≤ 3.5 Å, D-H···A angle ≥ 150°). A bond above about 30% is a persistent interaction worth describing; a few per cent is noise.

Avibactam

Residue	Occupancy (%)
SER130	62.5
SER235	45.7
ASN132	35.3
ARG244	28.6
SER70	27.6
ALA237	16.6
LYS234	2.4
GLU166	1.6
ARG277	0.3
ASN276	0.3
GLN278	0.2
GLN39	0.1

BTL-004

Residue	Occupancy (%)
SER130	64.3
ARG244	41.8
SER235	33.5
TYR105	8.6
SER70	1.8
ASN276	1.2
ASN132	1.1
VAL216	0.5
LYS234	0.3
MET129	0.2
GLY236	0.1
ALA237	0.1

BTL-007

Residue	Occupancy (%)
ARG222	6.8
ASN154	3.3
SER223	1.9
ARG277	0.5
LYS215	0.5

GLU274	0.3
TYR105	0.2
ARG191	0.2
GLN278	0.2
GLU197	0.2
LEU198	0.2
ASP38	0.2

BTL-023

Residue	Occupancy (%)
ASN132	6.2
GLY92	0.8
THR118	0.6
ARG244	0.2
ARG94	0.0
ARG120	0.0

Clavulanic_acid

Residue	Occupancy (%)
SER70	83.6
SER235	55.3
ALA237	50.5
SER130	33.0
ASN170	9.4
LYS234	1.9
MET129	0.3
TRP229	0.2
VAL216	0.2
LYS215	0.2
THR128	0.2
GLN39	0.2

Sulbactam

Residue	Occupancy (%)
ARG244	30.2
ASN276	18.0
SER235	13.1
ASN132	7.8
SER130	7.3
ALA237	3.5
TYR105	1.3
SER70	0.4

ARG277	0.3
LYS288	0.3
ARG275	0.2
GLN278	0.2

Limitations

Non-covalent treatment of the clinical inhibitors. Avibactam, clavulanic acid and sulbactam inhibit TEM-1 by covalent acylation of Ser70. Classical MD cannot form or break that bond, so their simulated residence times understate reality. They are included as non-covalent binders so that the comparison with the designed compounds is like for like.

Single replicate. One trajectory per compound. Differences between compounds that are small relative to the fluctuation within a single run should not be over-interpreted.

Protonation. States were assigned for pH 7.4 by functional group: carboxylic acids, sulfate esters and tetrazoles deprotonated, everything else neutral. No constant-pH simulation was performed.

Binding free energy. MM-PBSA / MM-GBSA was not computed in this run.